

Empirical Industrial Organization

Chapter 5 Problem Set

Merger Analysis

What this packet contains

This PDF is a worked problem set built from the Chapter 5 lecture material on merger analysis. It is designed as a revision packet: each problem is followed immediately by a detailed solution. The set covers:

- motivations for merger analysis in empirical IO,
- identification of conduct in homogeneous- and differentiated-product markets,
- unilateral and coordinated merger effects,
- ownership matrices and markup recovery,
- post-merger equilibrium simulation, and
- consumer welfare under mixed logit.

Study note

Notation follows the lecture wherever practical. Some exercises use slightly simplified symbols so that the algebra is easier to read on the page.

Contents

Quick reference	2
1 Foundations and conduct identification	3
2 Differentiated-products conduct	8
3 Merger effects and simulation mechanics	9
4 Welfare and coordinated effects	13
5 Capstone synthesis	16

Quick reference

Core notation

Q	Aggregate quantity in a homogeneous-product market.
q_f	Output of firm f .
$P(Q)$	Inverse demand function.
$MC_f(q_f)$	Marginal cost of firm f .
X	Demand shifter excluded from the pricing equation.
W	Cost shifter excluded from inverse demand.
Z	Demand rotator that changes the slope of inverse demand.
λ	Conduct parameter in the homogeneous-product model.
Δ	Ownership or conduct matrix in differentiated-products pricing.
$\Omega(p)$	Markup matrix built from demand derivatives and ownership.
$s(p)$	Vector of market shares.
mc	Vector of product-level marginal costs.

Conduct benchmarks

Market structure	Value of λ	Interpretation
Bertrand	0	A firm does not internalize the industry price effect of extra output.
Cournot	s_f	A firm internalizes the effect of its own quantity share on market price.
Full collusion	1	Firms act like a monopolist and internalize the full industry effect.

Five-step merger simulation workflow

1. Estimate demand and recover substitution patterns.
2. Use the pre-merger first-order conditions to back out marginal costs.
3. Replace the pre-merger ownership matrix Δ^{pre} with Δ^{post} .
4. Solve the post-merger equilibrium prices.
5. Compute prices, shares, markups, profits, and welfare under the counterfactual.

1 Foundations and conduct identification

Problem 1. Why merger analysis matters

Question

Why is merger analysis a central topic in empirical industrial organization? In your answer, explain what a merger simulation is trying to predict and why demand, cost, and conduct all matter.

Detailed answer

Merger analysis is central because antitrust authorities care about how a horizontal merger changes market power and market outcomes. A merger simulation is a counterfactual exercise: holding the maintained model fixed, we ask what equilibrium prices, quantities, shares, markups, profits, and consumer welfare would look like if the ownership structure changed. To answer that question, we need three building blocks. Demand determines substitution patterns and price elasticities. Cost determines the marginal cost side of firms' pricing incentives. Conduct determines how firms compete and what profits they internalize when they set prices or quantities. Without all three pieces, the merger counterfactual is incomplete or potentially misleading.

Problem 2. Interpreting the conduct parameter in homogeneous-product markets

Question

The chapter introduces the marginal revenue function

$$MR(Q) = P(Q) + \lambda QP'(Q).$$

Explain the economic meaning of the conduct parameter λ . What values of λ correspond to Bertrand competition, Cournot competition, and full collusion?

Detailed answer

The parameter λ summarizes how much of the industry-wide price effect a firm internalizes when it expands output. In a homogeneous-product Bertrand setting, a single firm cannot profitably move market price by changing its own output, so the relevant marginal revenue is just $P(Q)$, which corresponds to $\lambda = 0$. In a Cournot setting, firm f internalizes only the effect of its own quantity on market price, so $\lambda = s_f = q_f/Q$, the firm's quantity share. Under full collusion, firms behave like a single monopolist and internalize the entire industry price effect, so $\lambda = 1$. Thus, larger λ means more market power and a stronger markup motive.

Problem 3. A numerical example for marginal revenue**Question**

Suppose inverse demand is

$$P(Q) = 100 - 2Q.$$

At aggregate output $Q = 20$, compute $MR(Q)$ under: (i) Bertrand competition, (ii) Cournot competition for a firm with market share $s_f = 0.25$, and (iii) full collusion. Briefly interpret the ranking.

Detailed answer

Here $P(20) = 100 - 40 = 60$ and $P'(Q) = -2$. Therefore

$$MR(Q) = P(Q) + \lambda Q P'(Q) = 60 + \lambda \cdot 20 \cdot (-2) = 60 - 40\lambda.$$

So:

$$\text{Bertrand } (\lambda = 0) : MR = 60,$$

$$\text{Cournot with } s_f = 0.25 : MR = 60 - 40(0.25) = 50,$$

$$\text{Collusion } (\lambda = 1) : MR = 60 - 40 = 20.$$

The ranking is $60 > 50 > 20$. The more conduct approaches collusion, the more strongly the firm internalizes the price-reducing effect of extra output, so marginal revenue falls.

Problem 4. Deriving the first-order condition with conduct**Question**

Let inverse demand be $P(Q) = a - bQ$ with $b > 0$, and let firm f 's marginal cost be

$$MC_f(q_f) = c + dq_f,$$

where $d \geq 0$. Using the conduct parameter λ , derive the firm's first-order condition and explain how market power enters the equation.

Detailed answer

Because $P'(Q) = -b$, the general conduct-based first-order condition is

$$P(Q) + \lambda Q P'(Q) = MC_f(q_f).$$

Substituting in the functional forms gives

$$a - bQ + \lambda Q(-b) = c + dq_f,$$

or equivalently

$$a - bQ - \lambda bQ = c + dq_f.$$

The term λbQ is the markup force coming from internalizing the effect of extra industry output on price. When λ is larger, the firm behaves more strategically and is willing to restrict output more aggressively relative to a competitive benchmark.

Problem 5. Deriving the aggregate pricing equation**Question**

Suppose inverse demand is

$$P = a + bQ + dX + u_D, \quad b < 0,$$

and marginal cost for firm f is

$$MC_f = c + mq_f + hW + u_S.$$

Assume symmetry so that $q_f = Q/N$. Derive the aggregate pricing equation implied by the conduct first-order condition.

Detailed answer

The first-order condition is

$$P + \lambda Q P'(Q) = MC_f.$$

Since $P'(Q) = b$, symmetry implies

$$P + \lambda b Q = c + m \frac{Q}{N} + hW + u_S.$$

Rearranging,

$$P = c + \left(\frac{m}{N} - \lambda b \right) Q + hW + u_S.$$

This is the aggregate pricing equation. The slope on output,

$$\gamma = \frac{m}{N} - \lambda b,$$

combines the slope of marginal cost and the conduct parameter. That is exactly why reduced-form estimation alone does not generally recover conduct.

Problem 6. Why demand and cost shifters identify the reduced forms**Question**

Why can an excluded demand shifter X be used to instrument for Q in the pricing equation, and why can an excluded cost shifter W be used to instrument for Q in the inverse demand equation?

Detailed answer

Price and quantity are jointly determined in equilibrium, so Q is endogenous in both the demand and pricing equations. A valid demand shifter X affects demand but not marginal cost, so it moves equilibrium quantity through demand while remaining excluded from the supply-side error term. That makes it a candidate instrument for Q in the pricing equation. Symmetrically, a valid cost shifter W affects marginal cost but not demand, so it shifts quantity through supply while remaining excluded from the demand-side error term. With both types of excluded shifters, the

two reduced-form equations can be identified even though price and quantity are simultaneous outcomes.

Problem 7. Why conduct is not identified from reduced forms alone

Question

Using the aggregate pricing equation

$$P = c + \gamma Q + hW + u_S, \quad \gamma = \frac{m}{N} - \lambda b,$$

explain why the conduct parameter λ is not separately identified from reduced-form estimates of demand and supply.

Detailed answer

Reduced-form estimation identifies γ and the demand slope b , but γ is a combination of two structural objects: the slope of marginal cost, m/N , and the conduct term, $-\lambda b$. If we only know γ and b , then many pairs (m, λ) can generate the same observed pricing relation. Economically, a high price at a given quantity can be explained either by steep marginal cost or by a large markup due to strong market power. Without additional information, the data do not tell us which explanation is correct.

Problem 8. Recovering conduct when cost data are available

Question

Suppose the demand slope is $b = -2$, the marginal cost slope parameter is $m = 4$, the number of symmetric firms is $N = 4$, and the estimated pricing-equation slope is $\gamma = 1.5$. Using

$$\gamma = \frac{m}{N} - \lambda b,$$

recover λ and interpret it.

Detailed answer

Here

$$\frac{m}{N} = \frac{4}{4} = 1.$$

So the pricing equation becomes

$$1.5 = 1 - \lambda(-2) = 1 + 2\lambda.$$

Hence

$$2\lambda = 0.5 \quad \Rightarrow \quad \lambda = 0.25.$$

A conduct value of 0.25 is exactly what one would see in a symmetric four-firm Cournot market, since each firm's share would be $1/4$. So this estimate is consistent with Cournot-type behavior in this simple example.

Problem 9. Why a demand rotator is needed without cost data**Question**

Suppose cost data are not available, so the slope of marginal cost is unknown. What kind of extra variable is needed to identify conduct, and what does it do geometrically to demand?

Detailed answer

We need a *demand rotator*: a variable that changes the slope of inverse demand, not just its level. Geometrically, it rotates the inverse demand curve around some point rather than shifting the entire curve in parallel. This matters because conduct affects pricing through the term $QP'(Q)$. If we can observe exogenous variation in $P'(Q)$ while holding the underlying cost relation fixed, then movements in prices and quantities reveal how strongly firms internalize the price effect of output. Without such slope variation, conduct and marginal cost remain confounded.

Problem 10. Deriving the pricing equation with a demand rotator**Question**

Let inverse demand be

$$P(Q, Z) = a + bQ + \rho QZ + dX + u_D, \quad b < 0,$$

and let marginal cost be

$$MC_f = c + mq_f + hW + u_S.$$

Assume symmetry so that $q_f = Q/N$. Derive the pricing equation and show how λ is identified from the coefficient on QZ .

Detailed answer

Because

$$P'(Q) = b + \rho Z,$$

the conduct first-order condition is

$$P + \lambda Q(b + \rho Z) = c + m \frac{Q}{N} + hW + u_S.$$

Rearranging,

$$P = c + \left(\frac{m}{N} - \lambda b \right) Q - \lambda \rho QZ + hW + u_S.$$

So the coefficient on QZ in the pricing equation is

$$\gamma_2 = -\lambda \rho.$$

If the demand system separately identifies ρ , then

$$\lambda = -\frac{\gamma_2}{\rho}.$$

This is the key insight: the rotator creates a reduced-form term whose coefficient loads directly on conduct.

Problem 11. A numerical demand-rotator exercise**Question**

Suppose the inverse demand equation identifies $\rho = -0.5$, and the pricing equation identifies the coefficient on QZ as $\gamma_2 = 0.20$. Using

$$\gamma_2 = -\lambda\rho,$$

compute λ .

Detailed answer

Plugging into the identifying relation gives

$$0.20 = -\lambda(-0.5) = 0.5\lambda.$$

Hence

$$\lambda = \frac{0.20}{0.5} = 0.40.$$

So firms internalize 40 percent of the industry price effect in this reduced-form representation. That is more market-power-intensive than Bertrand and more collusive than symmetric Cournot with four firms, but less than full collusion.

2 Differentiated-products conduct**Problem 12. Conduct in differentiated-product markets****Question**

In the two-product differentiated-products first-order conditions, the chapter introduces parameters such as Δ_{12} and Δ_{21} . What do these parameters mean? What is the interpretation of $\Delta_{12} = \Delta_{21} = 0$ and of $\Delta_{12} = \Delta_{21} = 1$?

Detailed answer

These Δ -terms indicate whether a firm pricing product 1 internalizes the profit effect that its price has on product 2, and vice versa. If $\Delta_{12} = \Delta_{21} = 0$, each product is priced independently, so the firms behave like separate single-product competitors. If $\Delta_{12} = \Delta_{21} = 1$, the two products are priced jointly, as under common ownership or a fully internalized joint-pricing problem. More generally, intermediate or asymmetric values represent partial internalization, which can arise in models of partial coordination, bargaining, or misspecified ownership structures.

Problem 13. What variation identifies differentiated-products conduct**Question**

For differentiated products, why is it not enough to know only own-price elasticities? What additional variation is needed to identify terms like Δ_{12} and Δ_{21} ?

Detailed answer

To identify whether firms jointly internalize substitution across products, we need variation in the *cross-price* effects, such as $\partial Q_2/\partial p_1$ and $\partial Q_1/\partial p_2$. Own-price elasticities alone tell us how each product responds to its own price, but joint pricing depends on how raising the price of one product diverts consumers toward the other product. Therefore, identification requires some exogenous source of variation that rotates or shifts those cross-substitution terms while keeping the rest of the environment sufficiently stable. Intuitively, we must observe how the strength of diversion changes, not just how total demand shifts.

Problem 14. Why merger models must fit the institution**Question**

The chapter emphasizes that every market institution needs its own merger model. Why is that true? Give two examples from the notes.

Detailed answer

A merger model must match the institutional details of how firms actually compete. If the maintained model is wrong, the counterfactual can be badly misleading even if the econometrics are flawless. For example, hospitals and managed care organizations may negotiate both prices and coinsurance rates, so a simple posted-price Bertrand model is inappropriate. In supermarket retailing, the same chain may operate multiple stores, so competition is shaped by multi-store ownership rather than by single-store firms. The chapter also warns that in some industries firms may compete in advertising or use dynamic pricing, which again changes the correct equilibrium concept.

3 Merger effects and simulation mechanics**Problem 15. Unilateral, multilateral, and coordinated effects****Question**

Distinguish clearly between unilateral effects, multilateral effects, and coordinated effects of a horizontal merger.

Detailed answer

A *unilateral effect* is the direct incentive of the merged firm to raise price because it now internalizes the business that would be diverted from one of its products to another product it also owns. A *multilateral effect* is the broader equilibrium chain reaction: once the merged firm raises price, rivals may also find it optimal to adjust upward, and their reactions feed back into the final equilibrium. A *coordinated effect* is different. It means the merger changes the mode of competition itself, for example by making tacit collusion easier to sustain. In that case, the post-merger conduct parameters can change, not just the ownership matrix.

Problem 16. Writing the multiproduct firm's optimization problem**Question**

Suppose firm f owns the set of products J_f . Let demand for product j be $D_j(p)$, and let marginal cost be constant at mc_j . Write the firm's profit-maximization problem and derive the first-order condition for a product $k \in J_f$.

Detailed answer

The firm's problem is

$$\max_{p_f} \sum_{j \in J_f} (p_j - mc_j) D_j(p_f, p_{-f}).$$

Differentiate with respect to p_k , where $k \in J_f$:

$$\frac{\partial \Pi_f}{\partial p_k} = D_k(p) + \sum_{j \in J_f} (p_j - mc_j) \frac{\partial D_j(p)}{\partial p_k} = 0.$$

The first term is the direct gain from a higher price on product k . The summation term captures how changing p_k affects the demands of all products owned by the firm, including diversion toward other products in the portfolio.

Problem 17. Encoding ownership with the matrix Δ **Question**

Define the ownership matrix Δ^{pre} for a pre-merger market and use it to rewrite the first-order conditions in product-level form.

Detailed answer

Define

$$\Delta_{jk}^{pre} = \begin{cases} 1 & \text{if products } j \text{ and } k \text{ are owned by the same firm before the merger,} \\ 0 & \text{otherwise.} \end{cases}$$

Then the first-order condition for any product k can be written as

$$D_k(p) + \sum_{j \in J} \Delta_{jk}^{pre} (p_j - mc_j) \frac{\partial D_j(p)}{\partial p_k} = 0.$$

Using shares instead of demands gives the same structure:

$$s_k(p) + \sum_{j \in J} \Delta_{jk}^{pre} (p_j - mc_j) \frac{\partial s_j(p)}{\partial p_k} = 0.$$

This representation is powerful because a merger simulation often amounts to changing only the ownership matrix from Δ^{pre} to Δ^{post} .

Problem 18. Recovering marginal costs from the pre-merger equilibrium

Question

Define the matrix

$$\Omega^{pre}(p)_{jk} = -\Delta_{jk}^{pre} \frac{\partial s_j(p)}{\partial p_k}.$$

Show how pre-merger marginal costs can be recovered from observed prices and estimated demand derivatives.

Detailed answer

Stacking the first-order conditions across products yields

$$s(p) - \Omega^{pre}(p)(p - mc) = 0,$$

where $s(p)$ is the vector of market shares and mc is the vector of marginal costs. Rearranging gives

$$p - mc = [\Omega^{pre}(p)]^{-1} s(p).$$

Therefore,

$$mc = p - [\Omega^{pre}(p)]^{-1} s(p).$$

The term $[\Omega^{pre}(p)]^{-1} s(p)$ is the markup vector implied by the observed pre-merger equilibrium and the estimated demand system.

Problem 19. A numerical pre-merger cost recovery exercise

Question

Consider two products that are owned separately before the merger, so

$$\Omega^{pre} = \begin{pmatrix} 0.08 & 0 \\ 0 & 0.05 \end{pmatrix}.$$

Suppose observed prices are $p = (4, 5)'$ and observed shares are $s = (0.20, 0.15)'$. Recover the

marginal cost vector mc .

Detailed answer

Because the matrix is diagonal, its inverse is

$$(\Omega^{pre})^{-1} = \begin{pmatrix} 12.5 & 0 \\ 0 & 20 \end{pmatrix}.$$

Hence the markup vector is

$$(\Omega^{pre})^{-1}s = \begin{pmatrix} 12.5 & 0 \\ 0 & 20 \end{pmatrix} \begin{pmatrix} 0.20 \\ 0.15 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 3.0 \end{pmatrix}.$$

Therefore

$$mc = p - (\Omega^{pre})^{-1}s = \begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 2.5 \\ 3.0 \end{pmatrix} = \begin{pmatrix} 1.5 \\ 2.0 \end{pmatrix}.$$

So the implied marginal costs are 1.5 for product 1 and 2.0 for product 2.

Problem 20. A one-step post-merger price approximation

Question

Now suppose the same two products merge, so the post-merger matrix is approximately

$$\Omega^{post} = \begin{pmatrix} 0.08 & -0.02 \\ -0.015 & 0.05 \end{pmatrix}.$$

Assume marginal costs stay at $mc = (1.5, 2.0)'$ and, as a rough one-step approximation, the current shares remain $s = (0.20, 0.15)'$. Use

$$p^{post} \approx mc + (\Omega^{post})^{-1}s$$

to compute approximate post-merger prices. Compare them with the pre-merger prices $(4, 5)'$.

Detailed answer

First invert the matrix:

$$(\Omega^{post})^{-1}s \approx \begin{pmatrix} 3.514 \\ 4.054 \end{pmatrix}.$$

Therefore the approximate post-merger prices are

$$p^{post} \approx \begin{pmatrix} 1.5 \\ 2.0 \end{pmatrix} + \begin{pmatrix} 3.514 \\ 4.054 \end{pmatrix} = \begin{pmatrix} 5.014 \\ 6.054 \end{pmatrix}.$$

Compared with the pre-merger prices $(4, 5)'$, both products become roughly one dollar more expensive. This happens because the merged firm now internalizes substitution between the two

products, so the markup vector becomes larger. In a full simulation, shares and derivatives would also adjust, so this is only an approximation, but it captures the basic unilateral incentive.

Problem 21. Why the post-merger equilibrium is nonlinear

Question

The chapter writes the post-merger equilibrium as

$$p^{post} = mc + [\Omega^{post}(p^{post})]^{-1} s(p^{post}).$$

Why is this a nonlinear system, and why can it not usually be solved by one matrix inversion alone?

Detailed answer

The equation is nonlinear because both the share vector $s(\cdot)$ and the derivative-based matrix $\Omega^{post}(\cdot)$ depend on the very price vector we are trying to solve for. When prices change, market shares change; when shares change, substitution patterns and the demand Jacobian also change. That means the markup matrix itself must be recomputed at the candidate post-merger prices. So the merger problem is typically solved with a fixed-point algorithm or a nonlinear equation solver rather than with a single closed-form calculation.

Problem 22. The standard merger-simulation workflow

Question

Give a clean step-by-step description of a standard differentiated-products merger simulation under Bertrand-Nash pricing with separable marginal costs.

Detailed answer

A standard workflow has five steps. First, estimate the demand system and recover the substitution matrix, price elasticities, and market shares. Second, use the pre-merger ownership matrix Δ^{pre} together with the observed equilibrium to back out marginal costs from the first-order conditions. Third, replace Δ^{pre} with the counterfactual ownership matrix Δ^{post} implied by the merger. Fourth, solve the post-merger pricing fixed point

$$p^{post} = mc + [\Omega^{post}(p^{post})]^{-1} s(p^{post}).$$

Fifth, compute post-merger outcomes such as prices, shares, markups, profits, and consumer surplus. The entire exercise is valid only under the maintained assumptions about demand, cost, and conduct.

4 Welfare and coordinated effects

Problem 23. Why separable cost assumptions are convenient but restrictive**Question**

The notes often assume constant or separable marginal costs when simulating unilateral effects. Why is that assumption convenient, and what becomes more difficult when costs are nonseparable?

Detailed answer

With separable costs, the merger changes firms' pricing incentives without mechanically changing the marginal cost function. That is convenient because it lets the researcher isolate the ownership effect: prices change because products are jointly priced, not because production technology changes. If costs are nonseparable, however, a merger may create cost synergies, scope economies, or other interactions across products. Then the post-merger counterfactual no longer changes only the ownership matrix; it may also change marginal costs themselves. In that case, one must model how production costs change after integration, which is much more demanding both conceptually and empirically.

Problem 24. Consumer surplus in a mixed-logit model**Question**

Write the mixed-logit compensated-variation formula used in the chapter and explain what it measures.

Detailed answer

If the price coefficient for consumer i is $\alpha_i < 0$ and price enters utility linearly, the chapter's welfare measure is

$$CV_{it} = \frac{\ln\left(\sum_{j=0}^J \exp(V_{ijt}^{post})\right) - \ln\left(\sum_{j=0}^J \exp(V_{ijt}^{pre})\right)}{\alpha_i}.$$

Because $\alpha_i < 0$, a fall in the log-sum from pre to post generates a positive compensation amount. Economically, CV_{it} is the amount of money that would have to be given to consumer i after the merger-induced price change to make that consumer as well off as before. Aggregating this quantity across consumers gives a money-metric measure of the welfare effect.

Problem 25. A numerical compensated-variation exercise**Question**

Suppose one consumer has $\alpha_i = -2$. The inclusive-value term falls from 3.0 before the merger to 2.6 after the merger. Compute the consumer's compensated variation and interpret the result.

Detailed answer

Using the formula

$$CV_i = \frac{2.6 - 3.0}{-2} = \frac{-0.4}{-2} = 0.2.$$

So the consumer's compensated variation is 0.2 units of money. The positive sign means the merger made the consumer worse off, and the consumer would need 0.2 units of compensation to be brought back to the pre-merger level of utility.

Problem 26. Why linear price entry matters for welfare formulas**Question**

Why does the chapter emphasize that the log-sum consumer-surplus formula holds only when price enters utility linearly?

Detailed answer

The simple log-sum formula relies on the fact that the marginal utility of income is constant, which is what linear price entry implies in this setting. If income effects are present or the marginal utility of income varies with the level of consumption, then a single division by the price coefficient is no longer enough to convert utility changes into money changes. In that more general case, welfare calculations require additional structure or numerical integration. So linear price entry is not just a convenience; it is what makes the closed-form money-metric welfare formula valid.

Problem 27. Coordinated effects and repeated-game incentive constraints**Question**

How does repeated-game theory enter the analysis of coordinated effects after a merger? What must be checked for collusion to be sustainable?

Detailed answer

Repeated-game theory says that collusion is sustainable only if it is incentive compatible. For each firm, the present value of sticking to the collusive agreement must be at least as large as the one-time gain from deviating plus the continuation value after punishment. A merger can change this tradeoff by altering market shares, diversion patterns, markups, cost asymmetries, and the severity of future punishments. Therefore, coordinated-effects analysis asks whether the merger makes the incentive constraints easier or harder to satisfy. This is fundamentally different from unilateral-effects analysis, which usually holds the conduct regime fixed.

Problem 28. Why coordinated-effects analysis is harder than unilateral-effects analysis

Question

Why is prospective coordinated-effects analysis empirically harder than prospective unilateral-effects analysis?

Detailed answer

Unilateral-effects analysis can often proceed by holding the equilibrium concept fixed and changing only ownership. Coordinated-effects analysis is harder because it requires a model of how conduct itself changes after the merger. To quantify this, the researcher needs demand and cost estimates, but also a maintained theory of collusion or joint conduct and a way to identify the baseline conduct regime in the data. If conduct is unknown, then even cost recovery can become difficult because markups depend on conduct. For that reason, coordinated-effects work is much more assumption-sensitive and typically requires unusually rich institutional information.

5 Capstone synthesis

Problem 29. What information helped identify coordinated effects in Miller and Weinberg (2017)

Question

The chapter mentions Miller and Weinberg (2017) in connection with the MillerCoors joint venture. What empirical ingredients helped them learn about post-merger coordination?

Detailed answer

The key idea is that outsiders can help identify common cost shocks and separate them from changes in conduct. In the chapter's summary, unobserved year-specific and region-specific cost shocks are inferred using outsider products, while unobserved product-specific cost shocks are assumed to remain the same before and after the merger. Those assumptions allow the researcher to compare pre- and post-joint-venture outcomes and ask whether observed price changes are consistent with a simple ownership change or instead require a change in conduct. That is precisely the kind of structure needed to study coordinated effects empirically.

Problem 30. Why merger simulations are only as credible as their assumptions**Question**

The notes warn that merger simulations are valid only if the maintained model is correct. Explain why this warning is so important.

Detailed answer

A merger simulation is a structural counterfactual, not a purely descriptive extrapolation. It uses estimated primitives to predict what equilibrium would occur under a different ownership structure. If the model gets the nature of competition wrong - for example, if firms actually compete in advertising, negotiate contracts, or use dynamic pricing while the researcher assumes static posted-price competition - then the recovered marginal costs, markups, and counterfactual prices can all be wrong. This is why institutional fit matters so much in merger work: the counterfactual inherits the strengths and weaknesses of the model that produced it.

Problem 31. A capstone design question for an antitrust report**Question**

Suppose you are advising an antitrust authority about a proposed merger in a differentiated-products market. What key outputs would you report, and what assumptions would you make explicit in the write-up?

Detailed answer

A useful antitrust report would present at least the predicted post-merger prices, market shares, markups, firm profits, and consumer-surplus change, along with sensitivity analyses. It would distinguish unilateral from coordinated effects and explain whether any estimated cost efficiencies were included. I would also report the main demand elasticities, diversion ratios, and the ownership structure used in the simulation, because those objects drive the merger results. On assumptions, I would be explicit about the demand specification, the treatment of marginal costs, the equilibrium concept (for example, Bertrand-Nash pricing), whether conduct is assumed or identified, and whether costs are separable. Finally, I would flag institutional features - negotiations, dynamic pricing, multi-store ownership, or entry responses - that might make the baseline model too narrow.